

A strict polynomial subordination inequality for two explicit prime-generated discrete measures

An open problem, stated self-contained

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Abstract

We define, completely explicitly, a family of pairs (μ, ν) of finite positive discrete measures on $[-1, 1]$, built from the von Mangoldt function through a fixed elementary sampling recipe. Writing S for the total number of atoms of the pair, the problem is to prove or refute:

$$\int p^2 d\nu < \int p^2 d\mu \quad \text{for every real polynomial } p \neq 0 \text{ with } \deg p < \frac{S+1}{2}.$$

Numerically the inequality holds on all 42 instances of the family that we have tested, with a spectral margin that shrinks from $1.7 \cdot 10^{-4}$ to $1.4 \cdot 10^{-7}$ as S grows from 367 to 1755, and it genuinely fails a few degrees beyond the stated bound. Structurally destroyed comparison inputs lose the inequality already at 11–15% of the critical degree. The degree bound $(S+1)/2$ is exactly the moment-freedom count of an S -atom signed measure. No proof is known; a counterexample at large S would be equally valuable. This document is a problem statement, not a claim.

Status note (August 27, 2026). The research programme that produced this problem has paused its internal work on it and frozen it as a specialist problem. The new Section 6 carries the current state: the measurement campaign since the update of §3 has compressed the problem into *one object* — an almost perfect joint wander of two fine structures whose difference is five hundred times smaller than either — and three precise questions. Sections 1–5 are unchanged.

1 The problem

1.1 The construction

Let Λ denote the von Mangoldt function ($\Lambda(n) = \log q$ if $n = q^m$ is a prime power, else 0), and let γ be Euler’s constant. Fix once and for all the *oversampling parameter* $\nu_0 = 4$ and the table cap $T = 4 \cdot 10^5$ (all prime powers used below lie far below T ; the cap only delimits the admissible index list in §1.3).

Fix a prime power z (the *window anchor*) and let z' be the next larger prime power. Define

$$\alpha = \log z, \quad \delta = \frac{\log z' - \log z}{2\nu_0}, \quad M = \left\lceil \frac{\alpha}{\delta} - 10^{-9} \right\rceil + 1 \quad (\text{increased by 1 if odd}),$$

$$\Delta = \frac{2\alpha}{M}, \quad L = 2M - 2, \quad N = \frac{M}{2}.$$

Let $t_\Delta(x) = \max(0, 1 - |x|/\Delta)$ be the unit triangle (“tent”) of half-width Δ .

Step 1 (lag sequence). For $i = 0, 1, \dots, M - 1$ set $c_i = A(i\Delta) + P(i\Delta)$, where

$$P(s) = - \sum_{n \leq z^2} \frac{\Lambda(n)}{\sqrt{n}} \left[t_\Delta(s - \log n) + t_\Delta(s + \log n) \right], \quad (1)$$

$$A(s) = -(\gamma + \log \pi) t_\Delta(s) + \int_0^\infty \frac{2 t_\Delta(s) e^{-2w} - [t_\Delta(s - w) + t_\Delta(s + w)] e^{-w/2}}{1 - e^{-2w}} dw. \quad (2)$$

Both integrands are piecewise smooth and absolutely integrable (A is the archimedean term of the Weil explicit formula evaluated on the tent test function $t_\Delta(\cdot - s)$; P is the triangular-window sampling of the prime comb — but the two formulas above are the complete definition, no outside input is needed).

Step 2 (spectral density of the symmetrized lags). Define the even trigonometric polynomial

$$f(\theta) = c_0 + 2 \sum_{k=1}^{M-2} c_k \cos(k\theta) + c_{M-1} \cos((M-1)\theta), \quad (3)$$

i.e. the discrete Fourier transform of the symmetric extension $(c_0, \dots, c_{M-1}, c_{M-2}, \dots, c_1)$ of the lag sequence; the values $d_j = f(\theta_j)$ at $\theta_j = 2\pi j/L$ are the eigenvalues of the circulant extension of the $L \times L$ symmetric Toeplitz matrix with symbol f .

Step 3 (folding to $[-1, 1]$ and sign split). For $j = 1, \dots, L/2$ put

$$x_j = \cos \theta_j, \quad \tilde{w}_j = \kappa_j \frac{2}{L} (1 - \cos \theta_j) f(\theta_j), \quad \kappa_j = \begin{cases} 1, & 1 \leq j < L/2, \\ \frac{1}{2}, & j = L/2. \end{cases} \quad (4)$$

Discard indices with $\tilde{w}_j = 0$ and let S be the number of remaining atoms. Define the two positive measures

$$\mu = \sum_{\tilde{w}_j > 0} \tilde{w}_j \delta_{x_j}, \quad \nu = \sum_{\tilde{w}_j < 0} (-\tilde{w}_j) \delta_{x_j}, \quad (5)$$

with atom counts S_+ and S_- , $S_+ + S_- = S$. Both are finite positive measures with finite (disjoint) supports in $[-1, 1)$.

1.2 The concrete running example ($z = 16$)

For $z = 16$ (so $z' = 17$, $\alpha = \log 16 = 2.772589\dots$, $\delta = \frac{1}{8} \log \frac{17}{16} = 0.00757808\dots$) the recipe gives

$$M = 368, \quad \Delta = 0.01506842\dots, \quad L = 734, \quad N = 184,$$

70 prime powers $n \leq 256$ enter (1), and the fold produces

$$S = 367, \quad S_+ = 263, \quad S_- = 104, \quad \mu(\mathbb{R}) = 3.699410\dots, \quad \nu(\mathbb{R}) = 0.226205\dots$$

(every one of the 367 candidate atoms carries a nonzero weight, so $N = (S + 1)/2$). Table 1 shows an excerpt; the companion script (§7) prints every atom.

1.3 The admissible family and the question

Call the anchor z *admissible* if $N = M/2 \in [128, 900]$ and at least 40 prime powers lie below z^2 (with z drawn from the prime powers below $3.8 \cdot 10^5$, well inside the cap T). Exactly 42 prime powers are admissible,

$$z \in \{16, 23, 25, 27, 29, 31, 37, 41, 43, 47, 49, 53, 59, 61, 64, 67, 71, 73, \\ 79, 81, 83, 89, 97, 103, 109, 113, 121, 128, 131, 139, 151, 157, 163, 169, \\ 173, 181, 199, 211, 233, 243, 293, 317\},$$

Table 1: Atom excerpt for $z = 16$ ($L = 734$; $x_j = \cos(2\pi j/L)$). Positive weights belong to μ , negative weights (in absolute value) to ν .

j	x_j	$f(\theta_j)$	$\tilde{w}_j$	channel
1	+0.999963	+27.427975	$+2.738 \cdot 10^{-6}$	μ
2	+0.999853	-10.075060	$-4.023 \cdot 10^{-6}$	ν
3	+0.999670	+4.824158	$+4.334 \cdot 10^{-6}$	μ
4	+0.999414	-2.729923	$-4.360 \cdot 10^{-6}$	ν
5	+0.999084	+1.709341	$+4.266 \cdot 10^{-6}$	μ
6	+0.998681	-1.136329	$-4.083 \cdot 10^{-6}$	ν
...				
183	+0.004280	+8.446382	$+2.292 \cdot 10^{-2}$	μ
184	-0.004280	-2.274690	$-6.225 \cdot 10^{-3}$	ν
185	-0.012840	+6.118735	$+1.689 \cdot 10^{-2}$	μ
...				
365	-0.999853	+4.611240	$+2.513 \cdot 10^{-2}$	μ
366	-0.999963	+6.747708	$+3.677 \cdot 10^{-2}$	μ
367	-1.000000	-1.223818	$-3.335 \cdot 10^{-3}$	ν

with S ranging over $[283, 1755]$ and N over $[142, 878]$. We call $\{(\mu_z, \nu_z)\}$ the *prime window family* and $N = (S + 1)/2$ the *critical degree* of an instance.

Problem 1 (half-filling subordination). *Prove or refute: for every admissible anchor z and every real polynomial $p \neq 0$ with $\deg p < \frac{S+1}{2}$,*

$$\int p^2 d\nu_z < \int p^2 d\mu_z. \quad (L^*)$$

Any of the following would be decisive progress: (a) a proof of (L^*) for the full family, or for the unbounded family with the cap $N \leq 900$ removed; (b) a structural proof for a single instance (say $z = 16$) that explains the survival *exactly up to* the critical degree by a mechanism rather than by numerical evaluation; (c) a counterexample — an admissible z (with or without the cap) and an explicit p violating (L^*) . Since each instance is a statement about finitely many explicitly computable real numbers, a certified interval-arithmetic evaluation also settles any single instance rigorously; the interest is in the mechanism that makes the bound $(S + 1)/2$ sharp, uniformly in the family.

2 Equivalent formulations

Throughout fix an instance and write μ, ν, S, N as above, and $y_1, \dots, y_{S-}$ with weights $v_k > 0$ for the atoms of ν .

(a) Hankel positive definiteness. Let $m_k = \int x^k d(\mu - \nu)$ and $H_n = (m_{i+j})_{i,j=0}^{n-1}$. For $p(x) = \sum_{i < n} a_i x^i$ one has $\int p^2 d(\mu - \nu) = a^\top H_n a$, so (L^*) is exactly

$$H_N \succ 0 \quad \text{with } N = \frac{S+1}{2},$$

positive definiteness of the half-filling Hankel matrix of the *signed* measure $\mu - \nu$. Equivalently (Jacobi), all leading principal minors $D_1, \dots, D_N$ of H_N are positive, i.e. the three-term recurrence of $\mu - \nu$ has positive “pivots” $h_0, \dots, h_{N-1}$ (quasi-definiteness through the window).

(b) Christoffel–Darboux kernel as a strict contraction. Let $(p_i)_{i \geq 0}$ be the orthonormal polynomials of the *positive* measure μ and $K_n(x, y) = \sum_{i < n} p_i(x)p_i(y)$ its reproducing (Christoffel–Darboux) kernel. Define the $S_- \times S_-$ symmetric matrix

$$E_n[k, l] = \sqrt{v_k v_l} K_n(y_k, y_l), \quad 1 \leq k, l \leq S_-.$$

Writing $p = \sum_{i < n} a_i p_i$ and $B_{ki} = \sqrt{v_k} p_i(y_k)$ one has $\int p^2 d\mu = \|a\|^2$ and $\int p^2 d\nu = \|Ba\|^2$, and $E_n = BB^\top$ has the same nonzero spectrum as $B^\top B$. Hence

$$(L^*) \iff \lambda_{\max}(E_N) < 1 :$$

the ν -dressed Christoffel–Darboux kernel of μ at the half-filling degree is a strict contraction. (Equivalently: the Fredholm determinant $\det(I - s E_N)$ has no zero on $s \in (0, 1]$.)

(c) Quadrature/sampling form. E_n is the Gram matrix, in $L^2(\mu)$, of the functions $\sqrt{v_k} K_n(y_k, \cdot)$; the problem asks whether the sampling functionals $p \mapsto \sqrt{v_k} p(y_k)$ at the ν atoms, viewed inside $L^2(\mu)$ -normalized polynomials of degree $< N$, form a strict Bessel system with bound 1. This places the problem in the language of Marcinkiewicz–Zygmund/frame inequalities for weighted node sets.

3 What is known numerically

All statements in this section are measurements (double precision, with high-precision arbitration of every sign decision near zero); none of them is a theorem.

1. **The inequality holds on all 42 instances.** The margins $1 - \lambda_{\max}(E_N)$ are all strictly positive and *decrease* with the instance size:

z	S	N	$1 - \lambda_{\max}(E_N)$
16	367	184	$1.68 \cdot 10^{-4}$
23	301	151	$7.65 \cdot 10^{-5}$
31	867	434	$1.32 \cdot 10^{-5}$
59	979	490	$2.09 \cdot 10^{-6}$
128	1677	839	$1.82 \cdot 10^{-7}$
233	1717	859	$1.42 \cdot 10^{-7}$ (family minimum)
317	1067	534	$1.72 \cdot 10^{-7}$

The smallest instance ($z = 16$) has the *largest* margin; over the tested range the margin loses roughly three orders of magnitude. Any asymptotic reading of the family should take this shrinkage seriously: the unbounded version of Problem 1 may fail at large S , and a counterexample would be as informative as a proof.

2. **At the critical degree the contraction is broadly tight.** For $z = 16$ the top of the spectrum of E_{184} is

$$0.99983248, \quad 0.99874, \quad 0.99597,$$

three eigenvalues above 0.995 (confirmed by an independent 60-digit recomputation): the near-saturation is carried by several directions, not by one isolated mode.

3. **The bound is sharp within $O(1)$ degrees.** On every instance, $\lambda_{\max}(E_n)$ passes 1 at most six degrees above N ; the offset of the first Hankel-pivot sign failure beyond N has the census distribution $\{0:18, 1:10, 2:6, 3:6, 4:1, 5:1\}$ over the 42 instances. For $z = 16$ the failure is immediate: $\lambda_{\max}(E_{185}) = 1.00003660 > 1$.

4. **Structurally destroyed inputs fail early.** Feeding the same pipeline (Steps 1–3 verbatim) with destroyed input data — (i) the prime-comb positions replaced by i.i.d. uniform points on $[0, 2\alpha]$ at the same masses, (ii) the von Mangoldt weights replaced by the Dirichlet-series coefficients of the Epstein zeta function of $x^2 + 5y^2$, (iii) the comb replaced by its smooth density, and (iv) a random pair-correlation model — produces pairs (μ, ν) that lose (L^*) already at degrees 21/25/27/25 respectively (at $N = 184$), i.e. at 11–15% of the critical degree. Whatever mechanism drives (L^*) through the full window, it is a property of the actual prime comb, not of the sampling recipe.
5. **Norm bounds are far too coarse.** On the $z = 16$ instance the Gershgorin/row-sum bound on E_n already exceeds 1 at $n = 21$, and the trace bound at $n = 10$; both fail more than 160 degrees short of the truth. A proof of (L^*) must exploit cancellation, not absolute values.

What is known (update, 2026-08-25)

Three measured facts sharpen the picture above.

(a) *The margin decay obeys a flattening power law and the family was extended without a counterexample.* The admissible family was extended by 15 further instances with critical degrees $N = 942.1218$ (about 40% beyond the largest instance in the table); *every* margin $1 - \lambda_{\max}(E_N)$ remains strictly positive (minimum $+1.8 \cdot 10^{-8}$), each sign decision arbitrated at 30 and 45 digits of working precision, so the inequality now holds on all 57 tested instances. The decay follows a power law margin $\sim S^{-\alpha}$ with $\alpha \approx 3.05$ that *flattens* at large S (split-sample exponents 3.67/2.07), and the first-failure offset beyond N stays $O(1)$ on the new instances (maximum +4). The margin also falls *slower* than the local per-degree growth rate of $\lambda_{\max}$ at the crossing — the decay is consistent with a fixed $O(1)$ degree offset at increasing eigenvalue density rather than with an approaching failure. The unbounded version of Problem 1 remains open, but the tested range shows no crossing tendency.

(b) *The problem is not diophantine.* One might suspect that (L^*) consumes the number-theoretic (linear-forms-in-logarithms) structure of the node positions $\log n$. It does not: replacing *every* position $\log n$ by a nearby rational multiple of the grid step (continued-fraction convergents, denominators up to 56801, position change at most $2.1 \cdot 10^{-9}$, weights and grid cells preserved bitwise) — which destroys every exact linear-form relation among the positions — reproduces every measured quantity of the $z = 16$ instance *identically* ($\lambda_{\max}(E_{184}) = 0.99983248 < 1 < 1.00003660 = \lambda_{\max}(E_{185})$, confirmed at 60 digits). Baker–Matveev-type nonresonance input is therefore not what the inequality runs on (a needs-analysis puts the required exponent about four orders of magnitude beyond what such bounds deliver — and the rational surrogate shows it is not needed at all on the tested scales).

(c) *The mechanism is metrically sharp.* What the inequality *does* depend on is the exact metric placement of the nodes at sub-grid precision: randomizing the fractional parts of the node positions within their grid cells (distribution preserved exactly) destroys (L^*) once the perturbation exceeds roughly $10^{-3}..3 \cdot 10^{-3}$ of the local node gap, and a plain jitter ladder predicts the loss quantitatively. Any proof therefore has to consume the fractional-part *profile* of the positions as a metric object at that precision — but, by (b), nothing finer.

4 Why the degree bound matters

The critical degree is exactly the *moment-freedom count*. An S -atom signed measure has exactly S free moments $m_0, \dots, m_{S-1}$: the map (weights) $\mapsto$ (moment prefix) is a Vandermonde bijection, and every m_k with $k \geq S$ is forced linearly through the recurrence whose coefficients are the node polynomial $\prod_j (x - x_j)$. The Hankel block H_n consumes $m_0, \dots, m_{2n-2}$, so

$n = N = (S + 1)/2$ (for odd S) is the largest Hankel block determined entirely by free data: (L^*) says the form stays positive definite *through the whole free range and no further is asked*. Heuristically, the atoms sit at $S \approx L/2$ consecutive points of the image of an equispaced L -point grid on the unit circle, and a polynomial of degree n in $x = \cos \theta$ is an even trigonometric polynomial of degree n ; degrees up to $(S + 1)/2 \approx L/4$ are therefore precisely the sub-Nyquist regime in which p^2 (of trigonometric degree $\leq 2n \approx L/2$) is just barely unable to localize on individual grid atoms. Two hard counting facts frame the problem from above: with p the node polynomial of μ one gets $\int p^2 d(\mu - \nu) = -\int p^2 d\nu < 0$, so positive definiteness *must* fail by degree S_+ ; and the number of negative pivots over the full recurrence is exactly S_- (Jacobi/Sylvester). The observed failure sits at half-filling, far below the pigeonhole ceiling S_+ — counting alone cannot decide the problem, the content is metric.

5 Context

The inequality arises in a research program on finite-window versions of Weil positivity: the lag sequence $c_i = A(i\Delta) + P(i\Delta)$ is the Weil explicit-formula pairing evaluated on triangular test functions, so f in (3) is an explicit-formula spectral symbol, and (L^*) is equivalent to the positive definiteness of the associated finite quadratic form through the maximal moment-free range. The reduction of that program's central open question to precisely the inequality (L^*) , together with the measurement record quoted in §3, is documented in the program paper shipped alongside this note ([rh/paper/rh_program.pdf](#) in the same repository). We state explicitly: nothing here claims progress on the Riemann hypothesis or on Weil positivity; (L^*) is an open, self-contained, finite-dimensional problem in the approximation theory of signed discrete measures, and both a proof and a counterexample would be valuable.

6 The frozen state (August 2026): one object

This section was added on August 27, 2026, when the research programme paused internal work on Problem 1 and froze it as a specialist problem. It compresses the sealed measurement campaign that followed the update of §3 (internal record rounds r342–r354; every number below is quoted from a sealed, machine-checked record on finite instances). Nothing in this section is a theorem about the family, and nothing claims the inequality: the content is a *reduction* of the open problem to one measured object and three questions.

6.1 The extended census and the frame limit

Under sealed extension rules (the recipe of §1.1 verbatim, the window cap $N \leq 900$ lifted, anchors kept under the domain cap $z^2 \leq 4 \cdot 10^5$) the family was extended in five tranches to 85 instances with critical degree N up to 7942. On *every* instance the margin $1 - \lambda_{\max}(E_N)$ is strictly positive (deepest tranche, $N = 6532..7942$: $1.1 \cdot 10^{-10}$ to $1.7 \cdot 10^{-9}$; every near-zero sign decision in the campaign was arbitrated at 30–45 digits). Two frame facts matter for what follows. (i) The admissible pool of the construction is *exhausted* at $N = 7942 = 10^{3.90}$: the census cannot be extended inside this document's frame. (ii) On the deep tranches the margin decay *flattens* against the power-law fit of the first 57 instances — the deep rows sit uniformly on the high side of the extrapolation (a sealed two-parameter running-exponent model puts the census-grade limit exponent near 1.65 where the 57-instance fit gives 3.33; census only, no law is claimed — but the drift is in the direction friendly to the inequality).

6.2 The pair reduction and the typed chain

Fix an instance and let y_1, y_2 be the two atoms of ν nearest to $x = 1$ (the shallow edge; on the $z = 16$ instance these are the atoms at $j = 2, 4$). Write

$$d_k = v_k K_N(y_k, y_k) = (\mathbf{E}_N)_{kk}, \quad c = \sqrt{v_1 v_2} K_N(y_1, y_2) = (\mathbf{E}_N)_{12},$$

$$p = 1 - d_1, \quad q = 1 - d_2, \quad r = 1 - \frac{c^2}{pq} \quad (\text{the pair reserve}).$$

The extremal direction of $\mathbf{E}_N$ concentrates on this pair uniformly along the family (pair mass 0.935–0.999, participation ratio 1.61–1.94 on all tested instances), and at the binding point the inequality is exactly the determinant condition $c^2 < pq$ — a *backward* Cauchy–Schwarz statement about the ν -dressed kernel at two hard-edge atoms. The chain of measured structure above this pair is now fully typed:

1. **Weights: closed-form dictionary.** The pair weights v_1, v_2 are predicted in closed form from the construction — archimedean layer through the digamma dictionary $F_A(\xi) = -\log \pi + \operatorname{Re} \psi(\frac{1}{4} + i\xi/2)$ (with an explicit truncation tail), prime layer through an exact two-cell tent formula — at median relative error below 10^{-4} (maximum $9 \cdot 10^{-4}$) across the family.
2. **Deficits: candidate laws.** Along the family the bare scalars decay as N^{-a} with robust-fit exponents $a_p = 0.754$, $a_q = 0.645$, $a_c = 0.697$. The first-hit rational candidates $a_p = \frac{3}{4}$, $a_q = \frac{2}{3}$ pass every sealed cleanliness clause, but a hit is not an identification: separating $\frac{3}{4}$ from 2×0.38 needs a family reaching $N \sim 10^{31.5}$, $\frac{2}{3}$ from $1 - 0.38$ needs $10^{8.5}$, and the nearest ambiguity of the chain needs $10^{5.4}$ — against the pool end $10^{3.90}$. The candidates are *unresolvable inside the frame*.
3. **Pinning: theorem grade.** Split $\mathbf{E}_N = \begin{bmatrix} A & C \\ C^\top & D \end{bmatrix}$ (pair vs. rest) and dress the pair block: $A' = A + C(I - D)^{-1}C^\top$ with entries d'_1, d'_2, c' , deficits p', q' , reserve r' , and pair margin $m'_2 = 1 - \lambda_{\max}(A')$. Then $m'_2(p' + q' - m'_2) = p'q' - c'^2 = p'q'r'$ exactly (both dressed eigenvalue deficits are roots of $t^2 - (p' + q')t + (p'q' - c'^2)$); in the top-2-truncated resolvent model m'_2 equals the full margin identically and $p'q' - c'^2$ equals $\text{margin}^2 \cdot g_{21}/\Delta t^2$, exact in rational arithmetic; live, $|m'_2/\text{margin} - 1| \leq 0.0605$ on all 75 fully dressed instances. Consequently every dressed scalar decays at the margin rate times an $O(1)$ geometry factor: writing δ_x for the extra decay of x'/x , the identity forces $\delta_x = \alpha - a_x$ (matched live at 0.033), where $\alpha = 3.332$ is the measured margin exponent.
4. **Exponent bookkeeping closes with one census member.** $\alpha = a_p + a_q + \rho_r - a_{p+q} = 0.754 + 0.645 + 2.624 - 0.690 = 3.333$ against the measured 3.332 (deviation 0.001); with the rational candidates, $\frac{3}{4} + \frac{2}{3} + \rho_r - a_{p+q} = 3.352$ (deviation 0.019). Here $\rho_r = 2.624$, the decay exponent of the reserve r , is the single member of the chain that is neither dictionary nor candidate nor theorem: it is a census fit over a visibly curved family (halves curvature -0.767 ; no clean power law), and §6.3 identifies what it actually is.
5. **Computability closure.** The composition $c_{\text{pred}} = \sqrt{v_{\text{pred},1} v_{\text{pred},2}} K_{12}^{\text{cd}}$ — closed-form dictionary weights times the cross kernel evaluated by the Christoffel–Darboux recursion from the source measure, with no measured kernel input — reconstructs c on all 85 instances at maximum relative deviation $5.5 \cdot 10^{-4}$ and predicts the fine structure φ of §6.3 pointwise at correlation 1.0000000 and residual rms ratio 0.0004. This is a *computability* statement, not a closed form and not a derivation.

6.3 The one object: a shared wander that cancels

Set $A_{\text{lead}} = \frac{17}{12} = \frac{3}{4} + \frac{2}{3}$ (the sealed common leading exponent, the sum of the first-hit candidates) and define, across the family, the two fine structures

$$\varphi_D = \log(pq) + A_{\text{lead}} \ln N, \quad \varphi_K = \log(c^2) + A_{\text{lead}} \ln N.$$

The frozen finding (measured on the 81-instance census, reproduced exactly on the 85):

- φ_D and φ_K are *one shared wander*: correlation 0.999998, each with rms 0.88 nats, while their difference has rms 0.0017 nats — a factor ~ 500 smaller.
- The difference is not an approximation — it is pointwise the log reserve: $\varphi_D - \varphi_K = -\log(c^2/pq) = -\log(1 - r)$, and its decay refits ρ_r at $|\Delta\text{slope}| = 0.0002$. *The one unexplained exponent of the chain, $\rho_r = 2.624$, is exactly the decay of the difference of two near-identical wanders.*
- The wander has a measured anatomy. Via $c = \sqrt{v_1 v_2} K_N(y_1, y_2)$ and the Christoffel–Darboux endpoint formula, $\log c^2$ factorizes pointwise into four source blocks,

$$\log c^2 = W + B_2 + P_2 + G_M,$$

with weights $W = \log(v_1 v_2)$, recurrence prefactor $B_2 = 2 \log b_{N-1}$, CD polynomial $P_2 = 2 \log |p_N(y_1) p_{N-1}(y_2) - p_{N-1}(y_1) p_N(y_2)|$, and gap $G_M = -2 \log |y_1 - y_2|$. No single block carries φ : the pair-geometry block (P_2 , equivalently the composite $P_2 + G_M$) matches its *shape* at correlation +0.887 but with $2.4\times$ the amplitude, and the weight block W is *anti*-correlated at -0.72 and cancels the excess. φ is a two-block interference — a large pair-geometry wander almost cancelled by the dictionary-weight wander.

- φ is pair-local: it is not a function of N alone (instances of equal depth disagree at the full wander scale, adjacency ratio 0.911), and neither the local gap class nor the anchor size carries it (residual ratios 0.81/0.91).

The frozen question is therefore a single one: **why do the pair-geometry wander and the dictionary-weight wander cancel to a residual five hundred times smaller than either — equivalently, why does c track $\sqrt{pq}$ at two hard-edge atoms to this precision, with the cancellation decaying at the specific rate $\rho_r = 2.624$?**

6.4 Three questions for specialists (final form)

- (q1) **The backward Cauchy–Schwarz fine structure == the cancellation ratio.** For the ν -dressed Christoffel–Darboux kernel of μ at the two shallow-edge atoms, explain the near-saturation $c^2/pq \rightarrow 1$ at rate $N^{-2.624}$ — in the sharpest measured form: is the two-block cancellation ratio of §6.3 (pair geometry against dictionary weights, $2.4\times$ amplitude, correlation +0.887 against -0.72) a law?
- (q2) **The sub-classical Christoffel growth.** The within-window growth of $K_n(y, y)$ at the shallow-edge atoms has halves exponent ≈ 0.38 (census 0.336–0.608) — sub-classical for a hard edge — and the across-family deficit exponents hit the rational candidates $\frac{3}{4}, \frac{2}{3}$ and the 0.38-family ($2 \times 0.38, 1 - 0.38$) simultaneously. Deciding between them needs families reaching $N \sim 10^{5.4}$ (nearest ambiguity) up to $10^{31.5}$, while this construction ends at $10^{3.90}$. What is the true growth family, and why?

(q3) **The resolution paradox.** At the critical degree the kernel resolves the pair, and the bare reserve saturates ($r \rightarrow 0$ as $N^{-2.6}$); yet the *dressed* reserve r' stays $O(1)$ (span 0.138–0.536 over 75 instances, certified flat by a curvature-honest protocol; likewise the spectral gap ratio $g_{21} = (1 - \lambda_2)/(1 - \lambda_1)$, span 3.4–19.7). The dressing works by *signed cancellation*, $\Delta c \approx -c$: every absolute-value bound chain (triangle, Cauchy–Schwarz, operator norm) certifies none of the 75 instances, overshooting by factors 73 up to $2.4 \cdot 10^{11}$. Why does the dressing pin the dressed block at $O(1)$ — what controls the signed cancellation?

6.5 The honest frame

Everything above is a census on 85 finite instances with critical degrees up to $N = 7942$; no statement is asymptotic. All measured margins are strictly positive — no counterexample — and no proof mechanism is known; the inequality itself remains fully open, for the unbounded family in both directions. The candidate exponents are unresolvable inside this frame and the admissible pool is exhausted, which is precisely why the problem is frozen here rather than measured further: additional internal numerics on this family cannot decide anything that is still open. The record rounds cited above (r342–r354) are sealed, hash-pinned probe records in the research repository that ships this document; the companion script of §7 is unchanged and remains the machine check of the construction. This section, like the rest of the document, claims nothing: not the inequality, and nothing about the Riemann hypothesis.

6.6 Placement in the literature, and addressees

This closing subsection (added on the evening of August 27, 2026) translates the frozen object into the standard language of the communities whose toolkits contain it. It adds no new measurement and no claim; the classification below is census plus one clearly marked hypothesis.

Dictionary: the multi-point Uvarov transform

The signed measure $\mu - \nu$ is, verbatim, a multi-point (generalized) *Uvarov transform* of the positive measure μ : the transform subtracts the point masses v_k at the nodes y_k . Quasi-definiteness of such transforms is a classical subject. Kwon et al. (*Quasi-definiteness of generalized Uvarov transforms of moment functionals*, J. Appl. Math. 1 (2001)) characterize quasi-definiteness of the transformed functional by the nonvanishing of determinants $\det D_n \neq 0$ built from the Christoffel–Darboux kernel values $K_n(y_k, y_l)$ at the mass points; Aktaş–Area–Pérez (Comput. Appl. Math. 41 (2022), Thm 4) state the same criterion in matrix form: the transform is quasi-definite at degree n iff $I + \Lambda K_n$ is nonsingular, with Λ the diagonal matrix of the added masses and K_n the kernel matrix at the mass points. With $\Lambda = -\text{diag}(v_k)$ the matrix $I + \Lambda K_n$ is similar to $I - E_n$ of §2(b) (conjugate by $\text{diag}(\sqrt{v_k})$). The objects of this document are therefore exactly the standard ones: E_N is the multi-point Uvarov kernel matrix of the modification at the ν atoms; the free-window quasi-definiteness of §2(a) is the standard quasi-definiteness question for this transform, asked simultaneously at all degrees $n \leq N$; and the pair reserve $r = 1 - c^2/(pq)$ of §6.2 is a normalization of the 2×2 Uvarov determinant at the two binding atoms. What is *not* standard: the masses and nodes here are prime-generated rather than free parameters; the inequality asks the definite (spectral, $\lambda_{\max} < 1$) refinement of quasi-definiteness, uniformly through the moment-free window; and the frozen content is the measured *rate* at which the 2×2 determinant closes ($\rho_r = 2.624$).

Universality class: the discrete hard edge (census, one marked hypothesis)

Measured: the weight-free kernel correlation at the pair, $\rho_K = K_N(y_1, y_2)^2 / (K_N(y_1, y_1) K_N(y_2, y_2))$, *decays* along the family with fitted exponent 1.42 (record round r352; exactly twice the fitted exponent 0.711 of $\kappa = K_{12} / \sqrt{K_{11} K_{22}}$, the fit homogeneity exact). The two atoms sit at hard-edge-rescaled positions

$$a_j = 2N^2(1 - y_j), \quad a_j \rightarrow \frac{\pi^2 j^2}{4} \quad (j \in \{2, 4\} \text{ fixed, } N \rightarrow \infty),$$

asymptotically family-constant by the grid recipe ($y_j = \cos(2\pi j/L)$, $L = 4N - 2$); on the running instance $a_2 = 9.92$ and $a_4 = 39.69$, against the limits $\pi^2 = 9.87$ and $4\pi^2 = 39.48$. If μ belonged to the absolutely-continuous hard-edge universality class (Lubinsky, Int. Math. Res. Papers, 2008), the $2N^2$ -rescaled kernel would converge to the Bessel kernel, and ρ_K at these fixed rescaled positions would tend to a *positive constant*. It decays instead: the family is measurably outside the AC hard-edge class — consistent with the construction, since μ is purely atomic and the AC hypotheses do not apply. The edge lives in the *discrete* class. Within that class the natural regime is the saturation regime of Baik–Kriecherbauer–McLaughlin–Miller (*Discrete Orthogonal Polynomials: Asymptotics and Applications*, Ann. of Math. Studies 164, Princeton, 2007): the kernel is built from the orthonormal polynomials of the discrete measure μ used at degree $N \approx 0.70 S_+$ (on the running instance $184/263 = 0.6996$), and in that theory the edge parametrix adjacent to a saturated band is built out of Euler Gamma functions (Thm 2.7 and Remark 2.8 there) — noteworthy here because the closed-form weight dictionary of §6.2 is the digamma $\psi = (\log \Gamma)'$. Status of this placement: the decorrelation and the rescaled positions are *measured* and independently re-gated (round 356); the saturation-regime adjacency is a *hypothesis* (an internal round in progress probes the duality/regime side), and no asymptotic statement is claimed.

Who might recognize this

Four communities own pieces of this object. The discrete orthogonal-polynomial asymptotics school (the Baik–Kriecherbauer–McLaughlin–Miller line): Riemann–Hilbert asymptotics of discrete orthogonal polynomials, hard edges next to saturated bands — question (q2) and the class placement above live there. The Christoffel-function and universality community (the Lubinsky–Levin–Totik line): growth of $K_n(y, y)$ at edge atoms of purely discrete measures — (q2) is a Christoffel-function question, and the measured decorrelation is a universality-class datum. The Uvarov/spectral-transformation circle (the Marcellán school): E_N and the pair determinant are their standard objects (dictionary above); (q1) and (q3) are statements about multi-point Uvarov determinants and their dressed 2×2 blocks. The Weil-positivity environment (the Connes–Consani line): truncated and finite-rank forms of Weil positivity; the finite Guinand–Weil dictionary of arXiv:2607.02828 (2026) is a related finite tool through which the cancellation of §6.3 could be re-expressed. These are fields, not assignments: the paragraph names the schools whose standard toolkits contain the objects of this document, and it claims nothing on their behalf.

6.7 Addendum: the particle–hole reparametrization (round 356)

This closing addendum was added late on August 27, 2026. After the freeze, one further sealed internal round (record round r356, sealed and hash-pinned under the same conventions as §6.5) re-read the frozen object through the particle–hole duality of orthogonal polynomials on a finite set (A. Borodin, *Duality of orthogonal polynomials on a finite set*, J. Statist. Phys. 109 (2002), 1109–1120). It adds no claim and changes nothing in the frozen status; it sharpens the one-object question in three ways.

An exactly equivalent form: $R > \frac{1}{2}I$

Lift the pair to the *positive* measure $\eta = \mu + \nu$ with weights $u_j = |\tilde{w}_j|$ on the union support X . Two structural facts of the construction make Borodin’s duality apply verbatim, and both were checked bitwise on all 85 instances: X is the *full* folded cosine grid (no fold atom is ever culled, so $S = L/2$), and the critical degree satisfies exactly Borodin’s complementary rank condition $S = 2N - 1$ — the half-filling law $N = \lceil S/2 \rceil$ *is* the rank condition, with no off-by-one, across the whole family. Let $P_X(x) = \prod_{x' \in X} (x - x')$ be the monic node polynomial of X . Borodin’s dual (reciprocal) weight $u_j^\vee = 1/(u_j P'_X(x_j)^2)$ is, exactly on this grid,

$$u_j^\vee \propto \frac{\kappa_j (1 - x_j)}{|f(\theta_j)|}$$

(three independent evaluation routes agree: the grid closed form against the symbol form at $4.4 \cdot 10^{-16}$ relative, against the numeric node-polynomial route at $\leq 5.0 \cdot 10^{-9}$). Note the reciprocity: the original weight carries $+\log|f|$, the dual weight $-\log|f|$. Let $\Pi_{N-1}^{u^\vee}$ be the rank- $(N - 1)$ projection kernel of the orthonormal polynomials of $u^\vee$ on X , and let R be its restriction to $Y = \text{supp } \nu$ (dressed by $\sqrt{u^\vee}$, as $\mathbf{E}_n$ is by $\sqrt{v}$). Then, as finite-matrix identities (theorem grade; the inputs remain the measured window scalars): the L -ensemble transformation $Q = \mathbf{E}_N(I + \mathbf{E}_N)^{-1}$ (Q the Y -restriction of the rank- N projection kernel of η ; verified exactly in rational arithmetic on a small model, in double precision $\leq 7.6 \cdot 10^{-12}$ on all 85 instances) combined with Borodin’s complementation $\Pi_N^u = D(I - \Pi_{N-1}^{u^\vee})D$ at half filling (D the diagonal sign matrix of P'_X) gives $\mathbf{E}_N = D(R^{-1} - I)D$ on Y , hence

$$(L^*) \iff \lambda_{\max}(\mathbf{E}_N) < 1 \iff R > \frac{1}{2}I,$$

with the exact spectral map margin $= 2 - 1/\lambda_{\min}(R)$ (verified to $9.9 \cdot 10^{-10}$ absolute across the family), and the pair block of §6.2 is exactly the $(1, 2)$ principal 2×2 minor of $2I - R^{-1}$: $c = \varepsilon_1 \varepsilon_2 (R^{-1})_{12}$, $p = 2 - (R^{-1})_{11}$, $q = 2 - (R^{-1})_{22}$ (signs $\varepsilon_k = \pm 1$ from D ; verified to $4.5 \cdot 10^{-7}$ relative on the deepest instances). Live census: $\lambda_{\min}(R) - \frac{1}{2}$ ranges from $+4.2 \cdot 10^{-5}$ (the running instance) down to $+5.9 \cdot 10^{-11}$ (deepest tranche) — all positive, no counterexample, the census of §6.1 restated in hole coordinates; the structurally destroyed inputs of §3, item 4, violate $R > \frac{1}{2}I$ *structurally*, with $\lambda_{\min}(R) - \frac{1}{2} \in [-0.500, -0.083]$. Honest limit: at the deepest instances the double-precision resolution of the dual route ($\sim 5 \cdot 10^{-10}$) exceeds the margins ($\sim 4 \cdot 10^{-10}$), so the hole coordinate is exact algebra but not a sharper certification instrument.

The anti-correlation is by design

The $+\log|f| / -\log|f|$ reciprocity above makes the dual weight block the exact negative of the dictionary-weight block W of §6.3 up to a family-constant hard-edge geometry factor; gated live on the fine-structure operator across the family, the correlation of the two weight blocks is -0.999998 (sealed bar -0.99). The anti-correlated two-block interference of §6.3 — named there as part of the puzzle — is therefore *duality algebra*, not an independent phenomenon. This narrows the frozen question honestly: it is no longer “why do the pair-geometry and weight wanders couple with opposite sign” (they do so by algebra), but only “why is the difference of the by-design-coupled blocks five hundred times smaller than either block”.

The cancellation is locally encoded (census)

Let R_{pair} be the 2×2 restriction of R to the two binding atoms. Its inverse $(R_{\text{pair}})^{-1}$ — consuming R_{11}, R_{12}, R_{22} only: no readback of p, q, c and no global inversion — predicts the log reserve $-\log(1 - r) = -\log(c^2/pq)$ pointwise along the family: fine-structure correlation $+0.9982$ (leave-out residual 0.068) on the 57 calibration instances, and $+0.9828$ (leave-out

0.165) *pure-test* on the 28 deep instances never used in the fit; the pointwise log ratio has median -0.51 nats. The $\sim 500\times$ cancellation of §6.3 therefore lives in the *local dual two-point data* at the two atoms. This is a census-grade localization, not a mechanism. Equally honest: the duality itself does *not* compress the problem — no single dual block passes the sealed carrier clauses of the record (the round’s main verdict is “reparametrization only”), so the hole coordinate localizes the object but does not reduce it. Independently, the same round re-gated the class exclusion of §6.6: the rescaled pair positions are family-constant against $\pi^2 f^2/4$ on all 85 instances (maximum deviation $0.33/N$), and the decorrelation exponent 1.4222 of the earlier record is reproduced exactly — the AC/Bessel-class exclusion stands re-gated. Nothing in this addendum claims the inequality; the problem remains open exactly as stated in §6.5.

7 Verification script, contact, citation

The companion script `verify_lstar_instance.py` (same directory; `numpy/mpmath` only) rebuilds (μ, ν) for $z = 16$ from the formulas of §1.1 alone, cross-checks every atom position and weight against the research code that produced the measurement record (agreement to $4 \cdot 10^{-16}$ relative), reproduces $\lambda_{\max}(\mathbf{E}_{184}) = 0.99983248$ and $\lambda_{\max}(\mathbf{E}_{185}) = 1.00003660$ with a 60-digit ward, and verifies (L^*) directly on monomials and on 500 random polynomials of degree < 184 ; `--ladder` sweeps all 42 instances (the margin table of §3 is its output). It ends `LSTAR INSTANCE VERIFIED`.

Questions, proof sketches and counterexamples are equally welcome. Suggested citation: S. Hamann, *A strict polynomial subordination inequality for two explicit prime-generated discrete measures (open problem)*, TFPT research repository, `rh/problem/`, August 2026.